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Power conversion device with refrigerant

Abstract

This power conversion device comprises: first and second power circuit units each having a power semiconductor element and a plurality of conductors that hold the power semiconductor element therebetween and that are connected to an emitter and a collector of the power semiconductor element; and a flow channel forming body which houses the first and second power circuit units and through which a refrigerant flows. A conductor at the emitter side of the first power circuit unit is disposed so as to face a conductor at the collector side of the second power circuit unit. The conductor at the emitter side of the first power circuit unit and the conductor at the collector side of the second power circuit unit are connected to each other via a plurality of conductive fins which are in contact with the refrigerant.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0228757	12/2011	Kitami	257/713	H01L 23/36
2025/0079275	12/2024	Scholz	N/A	H01L 21/56

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007235060	12/2006	JP	N/A
2011064841	12/2010	WO	N/A

OTHER PUBLICATIONS

International Search Report of PCT/JP2021/039709 dated Dec. 28, 2021. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a power conversion device.

BACKGROUND ART

(2) In recent years, there is a demand for reduction in size and weight of the entirety of a vehicle with respect to an increase tendency of the amount of power converted by a power conversion device. Therefore, technological improvement for improving the output of the device while

suppressing an increase in size and weight of the power conversion device is performed every day. In addition, since an in-vehicle power conversion device is used in an environment in which a temperature change is large as compared with an industrial power conversion device or the like, a device capable of maintaining high reliability even in a high-temperature environment is required.
(3) When performing power conversion, the power conversion device needs a switching operation in which semiconductor modules constituting upper and lower arms of an inverter circuit repeat a cutoff state and a conduction state. At this time, the transient current flowing through the upper and lower arms is affected by the parasitic inductance of a wiring and becomes a cause of the surge voltage. As a result, the loss of the semiconductor module increases, and the temperature of a chip mounted therein increases. It is an important problem for a power conversion device having high reliability to achieve both reduction in inductance that causes the temperature rise and improvement in cooling performance for suppressing the temperature rise.
(4) PTL 1 as follows is known as the background art in the present invention. A cooling structure of a semiconductor device in PTL 1 includes two semiconductor elements that holds an output electrode and are disposed to face each other, and a radiator disposed on an opposite side of the output electrode with respect to the semiconductor elements. The output electrode includes an element mounting portion and a heat transport portion. The element mounting portion is electrically connected to the two semiconductor elements and is formed of a conductive material. The heat transport portion is provided to extend from the element mounting portion toward the radiator. With such a configuration, a technique of reducing the inductance that causes the temperature rise and having excellent cooling efficiency is disclosed.
CITATION LIST
Patent Literature
(5) PTL 1: International Publication No. 2011/064841
SUMMARY OF INVENTION
Technical Problem
(6) In the method disclosed in PTL 1, a plurality of input electrodes is arranged in parallel, and the parasitic inductance is canceled between the input electrodes, so that it is possible to reduce the switching loss. However, since the heat transport portion is connected to the radiator via an insulating member, there is a concern that the cooling efficiency of the insulating member on a heat radiation path is lowered. In view of such circumstances, an object of the present invention is to provide a power conversion device that achieves both improvement in cooling performance by both-side cooling excluding an insulating member on a heat radiation path and reduction in inductance for suppressing a switching loss, and increases an output.
Solution to Problem
(7) A power conversion device includes first and second power circuit units each including a power semiconductor element and a plurality of conductors that hold the power semiconductor element therebetween and that are connected to an emitter and a collector of the power semiconductor element, and a flow channel forming body which houses the first and second power circuit units and through which a refrigerant flows. The conductor at the emitter side of the first power circuit unit is disposed so as to face the conductor at the collector side of the second power circuit unit, and the conductor at the emitter side of the first power circuit unit and the conductor at the collector side of the second power circuit unit are connected to each other via a plurality of conductive fins which are in contact with the refrigerant.
Advantageous Effects of Invention
(8) According to the present invention, it is possible to provide a power conversion device that achieves both improvement in cooling performance and reduction in inductance, and increases an output.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a cross-sectional view of a power conversion device according to an embodiment of the present invention.

(2) FIG. **2** is a plan view of a board (lower layer) of a first power circuit unit according to the embodiment of the present invention.

(3) FIG. **3** is a plan view in which a molding resin is removed from FIG. **2**.

(4) FIG. **4** is a plan view of a power module.

(5) FIG. **5** is a perspective view of the power module.

(6) FIG. **6** is a front view of the power module.

(7) FIG. **7** is a plan view of a board (upper layer) of a second power circuit unit according to the embodiment of the present invention.

(8) FIG. **8** is a plan view in which the molding resin is removed from FIG. **7**.

(9) FIG. **9** is a top view of the power conversion device according to the embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

Embodiment of Present Invention

(10) Hereinafter, an embodiment of the present invention will be described with reference to the drawings. However, the present invention is not construed as being limited to the following embodiment, and the technical idea of the present invention may be realized by combining other known components. In the drawings, the same elements are denoted by the same reference signs, and repetitive description will be omitted.

(11) In addition, positions, sizes, shapes, ranges, and the like of the components illustrated in the drawings may not represent actual positions, sizes, shapes, ranges, and the like in order to facilitate understanding of the invention. Therefore, the present invention is not necessarily limited to the positions, sizes, shapes, ranges, and the like illustrated in the drawings.

(12) FIG. **1** is a cross-sectional view of a power conversion device according to an embodiment of the present invention.

(13) A power conversion device **100** is a power conversion device that converts DC power from a battery or the like into AC power and supplies the AC power to an electric motor. FIG. **1** illustrates a configuration of an upper arm circuit and a lower arm circuit for one phase.

(14) The power conversion device **100** includes a first board **3001** including a first power circuit unit **201**, a second board **3002** including a second power circuit unit **202**, a capacitor **40** that smooths a voltage applied to the power conversion device **100**, and a flow channel forming body **25** through which a refrigerant that cools the entirety of the power conversion device **100** flows.

(15) The first power circuit unit **201** connects an IGBT **10** that is a power semiconductor element, to a first power-circuit-side emitter conductor plate **221** and a first power-circuit-side collector conductor plate **211** by soldering or the like. Similarly, the second power circuit unit **202** connects the IGBT **10** that is a power semiconductor element, to a second power-circuit-side emitter conductor plate **222** and a second power-circuit-side collector conductor plate **212** by soldering or the like.

(16) The first power circuit unit **201** and the second power circuit unit **202** each include the IGBT **10**, the emitter conductor plate **221** (**222**), and the collector conductor plate **211** (**212**). In addition, the first power circuit unit **201** and the second power circuit unit **202** are installed in a power module assembly hole **303** formed to be installed in a board **30** (**3001**, **3002**) such as a printed circuit board. The first power circuit unit **201** and the second power circuit unit **202** installed on the board **30** (**3001**, **3002**) are sealed and fixed with a molding resin **23**. This eliminates the need for a bus bar having a complicated shape and improves productivity.

(17) The board **30** (**3001**, **3002**) includes a plurality of conductor layers made of a copper material or the like. Portions other than the conductor layers are made of an insulating member such as a glass epoxy resin, and conductors are formed in the respective layers via vias **302**. As a result, the cross-sectional area of the conductor is increased, and it is possible to reduce the inductance.
(18) As illustrated in FIG. **1**, the power conversion device **100** includes upper and lower two layers in which the first power circuit unit **201** serves as a lower layer and the second power circuit unit **202** servers as an upper layer. The flow channel forming body **25** is formed so as to cover the two power circuit units. Further, the flow channel forming body **25** includes three flow channel forming bodies of an upper-surface flow channel forming body **251**, a lower-surface flow channel forming body **252**, and an intermediate flow channel forming body **253**.
(19) The upper-surface flow channel forming body **251** forms a flow channel together with the second power circuit unit **202** and the second board **3002**. The refrigerant flowing in the upper-surface flow channel forming body **251** flows into the inside from a flow channel inlet **26**, and flows in communication with the intermediate flow channel forming body **253** through a through-hole **301***a*. As a result, the upper surfaces of the second power circuit unit **202** and the second board **3002** are cooled.
(20) The intermediate flow channel forming body **253** forms a flow channel together with the second power circuit unit **202** and the second board **3002**, and the first power circuit unit **201** and the first board **3001**. The refrigerant flowing in the intermediate flow channel forming body **253** flows into the intermediate flow channel forming body **253** from the upper-surface flow channel forming body **251** through a through-hole **301***a*, and then flows in communication with the lower-surface flow channel forming body **252** through a through-hole **301***b*. As a result, the lower surfaces of the second power circuit unit **202** and the second board **3002** and the upper surfaces of the first power circuit unit **201** and the first board **3001** are cooled.
(21) The lower-surface flow channel forming body **252** forms a flow channel together with the first power circuit unit **201** and the first board **3001**. The refrigerant flowing into the lower-surface flow channel forming body **252** flows into the lower-surface flow channel forming body **252** from the intermediate flow channel forming body **253** through the through-hole **301***b*, and then is discharged to the outside of the flow channel forming body **25** through a flow channel outlet **27**. As a result, the lower surfaces of the first power circuit unit **201** and the first board **3001** are cooled.
(22) With the configuration described with reference to FIG. **1**, a refrigerant path **254** is formed, and the refrigerant flows inside the flow channel forming body **25** along an arrow of the refrigerant path **254**. A cooling fin **24** is provided in the middle of the refrigerant path **254** to promote heat radiation effects of the first power circuit unit **201** and the second power circuit unit **202**.
(23) The cooling fin **24** has conductivity and electrically connects the first power-circuit-side emitter conductor plate **221** and the second power-circuit-side collector conductor plate **212**. As a result, it is possible to connect the power circuit units to each other at the shortest distance, which contributes to reduction in inductance. In addition, since it is not necessary to arrange the first power circuit unit **201** and the second power circuit unit **202** in parallel in a planar direction, this contributes to reduction in size of the power conversion device **100**.
(24) The capacitor **40** is installed outside the flow channel forming body **25**. A positive electrode terminal **401** is connected to the first board **3001**, and a negative electrode terminal **402** is connected to the second board **3002**. This prevents corrosion of an electronic component due to the contact of the refrigerant.
(25) FIG. **2** is a plan view of a board (lower layer) of the first power circuit unit according to the embodiment of the present invention. X-X′ is for indicating the cross-sectional position in FIG. **1**.
(26) The first board **3001** includes the first power circuit unit **201**, a positive electrode power supply terminal conductor **31** including a positive electrode power supply terminal **311**, an AC output terminal conductor **33** including an AC output terminal **331**, and a control circuit **50** that generates a control signal.

(27) The capacitor **40** is mounted between the first power circuit unit **201** and the positive electrode power supply terminal **311** on the first board **3001**. The capacitor **40** is configured by a ceramic capacitor or the like, and a plurality of capacitors are mounted. The capacitor **40** has a positive electrode terminal **401** and a negative electrode terminal **402** (FIG. **1**). The positive electrode terminal **401** is connected to the first board **3001**. Since the capacitor **40** is mounted in parallel with the first power circuit unit **201**, a current path from the capacitor **40** to the first power circuit unit **201** is expanded, so that it is possible to reduce the inductance.
(28) The first power circuit unit **201** is molded by the molding resin **23**, but a partial surface of the first power-circuit-side emitter conductor plate **221** is not molded and is exposed to a refrigerant flow channel. Further, a plurality of cooling fins **24** are formed on the exposed partial surface. Thus, the heat radiation effect of the first power circuit unit **201** is improved.
(29) FIG. **3** is a plan view in which the molding resin is removed from FIG. **2**.
(30) The control circuit **50** is disposed to be adjacent to the first power circuit unit **201**, and is connected to the first power circuit unit **201** from an in-flow channel board wiring **52** via a control signal wiring **51** such as wire bonding. As a result, the inductance of the control signal wiring **51** is reduced, and a decrease in element driving performance is prevented, thereby preventing an increase in loss.
(31) In the first board **3001**, the through-hole **301***b* through which the refrigerant passes is provided between the first power circuit unit **201** and the AC output terminal **331**. The through-hole **301***b* is formed in a circular shape or the like, and a plurality of through-holes are provided in parallel with the first power circuit unit **201**. The through-hole **301***b* causes the refrigerant to communicate with all the conductor layers provided on both surfaces of the board **3001**, and enables cooling of both surfaces of the first power circuit unit **201**.
(32) FIG. **4** is a plan view of the power module, FIG. **5** is a perspective view thereof, and FIG. **6** is a front view thereof.
(33) The power module **20** constitutes the upper arm circuit or the lower arm circuit for one phase in the power conversion device that converts DC power into AC power. The power module **20** includes an IGBT **10**, a diode **11**, a collector conductor plate **21**, and an emitter conductor plate **22**.
(34) The IGBT **10** includes a plate-shaped main electrode **101** and a control electrode **102** that controls a main current flowing through the main electrode **101**. The collector conductor plate **21** and the emitter conductor plate **22** are made of a copper material. The IGBT **10** and the diode **11** are sandwiched from both surfaces by the collector conductor plate **21** and the emitter conductor plate **22**. The IGBT **10** and the diode **11** are connected to the collector conductor plate **21** and the emitter conductor plate **22** via a metal bonding material **12** such as solder.
(35) In the power conversion device **100** in the present embodiment, each of the first power circuit unit **201** and the second power circuit unit **202** is configured by the power module **20** having the above-described structure. In the first power circuit unit **201**, the collector conductor plate **21** and the emitter conductor plate **22** correspond to the first power-circuit-side collector conductor plate **211** and the first power-circuit-side emitter conductor plate **221**, respectively. In the second power circuit unit **202**, the collector conductor plate **21** and the emitter conductor plate **22** correspond to the second power-circuit-side collector conductor plate **212** and the second power-circuit-side emitter conductor plate **222**, respectively. In FIG. **1**, the diode **11** is not illustrated.
(36) FIG. **7** is a plan view of a board (upper layer) of the second power circuit unit according to the embodiment of the present invention. X-X′ is for indicating the cross-sectional position in FIG. **1**.
(37) The second board **3002** includes the second power circuit unit **202**, a negative electrode power supply terminal conductor **32** including a negative electrode power supply terminal **321**, an AC output terminal conductor **33** including an AC output terminal **331**, and a control circuit **50** that generates a control signal. A plurality of through-holes **301***a* through which the refrigerant passes are provided between the second power circuit unit **202** and the negative electrode power supply terminal **321**, in parallel to the second power circuit unit **202**.

(38) Although the second board **3002** is formed with the same structure as the first board **3001**, the dispositions and the structures of the first board **3001** and the second board **3002** will be described in comparison with each other by using FIGS. **2** and **7**. The first board **3001** illustrated in FIG. **2** and the second board **3002** illustrated in FIG. **7** have point-symmetrical shapes except for the disposition of the capacitor **40**. That is, the positive electrode power supply terminal conductor **31** of the first board **3001** corresponds to the AC output terminal conductor **33** of the second board **3002**, and the AC output terminal conductor **33** of the first board **3001** corresponds to the negative electrode power supply terminal conductor **32** of the second board **3002**. As described above, since the upper and lower arms constituting the power conversion device **100** are manufactured with the same shape and structure, productivity is improved.
(39) Furthermore, as illustrated in FIG. **1**, by disposing the first board **3001** and the second board **3002** in an overlapping manner in the form of upper and lower layers, it is possible to simultaneously realize shortening of a wiring path and cooling by the refrigerant flowing in the flow channel forming body **25**. Therefore, this contributes not only to the effect of increasing the output by achieving both the improvement in the cooling performance and the reduction in the inductance, but also to the reduction in the size of the entire device.
(40) In the second board **3002**, the second power-circuit-side collector conductor plate **212** is disposed so as to face the first power-circuit-side emitter conductor plate **221**, and the first power-circuit-side emitter conductor plate **221** and the second power-circuit-side collector conductor plate **212** are connected to each other by the plurality of conductive cooling fins **24** (FIG. **1**). As a result, the disposition in which the IGBT **10** of the first power circuit unit **201** faces the diode **11** of the second power circuit unit **202**, and the diode **11** of the first power circuit unit **201** faces the IGBT **10** of the second power circuit unit **202** is obtained. With such a configuration, the transient current path at the switching time of the power conversion device **100** is shortened, and the counter currents flow through the first power circuit unit **201** and the second power circuit unit **202**, so that the inductance is reduced.
(41) The capacitor **40** is located between the second power circuit unit **202** and the negative electrode power supply terminal **321** on the second board **3002**. The negative electrode terminal **402** of the capacitor **40** is connected to the second board **3002**, and the negative electrode terminal **402** is disposed to face the positive electrode terminal **401** (FIG. **1**). As a result, a current path flowing out from the capacitor **40** to the first power circuit unit **201** faces a current path flowing into the capacitor **40** from the second power circuit unit **202**, and it is possible to reduce the inductance.
(42) Further, in the configuration described above, in the power conversion device **100**, electric energy necessary for driving the electric motor is supplied from the battery to the first power circuit unit **201** and the second power circuit unit **202**, and the AC power output from the AC output terminal **331** provided in the AC output terminal conductor **33** is controlled. At the switching time of the power conversion device **100**, the current flowing out of the positive electrode terminal **401** of the capacitor **40** flows from the first power circuit unit **201** into the second power circuit unit **202** via the conductive cooling fin **24**, and then flows into the negative electrode terminal **402**. As a result, the transient current path at the switching time is shortened, and the inductance is reduced.
(43) Furthermore, a heat radiation path is formed from the semiconductor elements (IGBT **10** and diode **11**) of the first power circuit unit **201** and the second power circuit unit **202** to the conductive cooling fins **24** without an insulating member interposed therebetween, and the heat radiation path is directly cooled by the refrigerant such as oil, so that it is possible to suppress an increase in thermal resistance and increase the output of the power conversion device **100**. In addition, by providing the through-holes **301***a* and **301***b* close to the molding resin **23**, it is possible to shorten the refrigerant path **254** and to achieve both the reduction in pressure loss and the improvement in cooling efficiency.
(44) FIG. **9** is a top view of the power conversion device according to the embodiment of the

present invention. X-X′ is for indicating the cross-sectional position in FIG. **1**.

(45) The flow channel forming body **25** is provided so as to surround the molding resin **23** and the through-holes **301***a* and **301***b*. The capacitor **40** and the control circuit **50** are formed outside the flow channel forming body **25**. This makes it possible to prevent corrosion of the electronic component due to the contact of the refrigerant. On the other hand, the first power circuit unit **201**, the second power circuit unit **202**, and the control signal wiring **51** are configured in a flow channel forming area, but are not electrically affected by the contact of the refrigerant by the molding resin **23**.

(46) According to the above-described embodiment of the present invention, the advantageous effects as follows are exhibited.

(47) (1) A power conversion device **100** includes: first and second power circuit units **201** and **202** each including an IGBT **10** that is a power semiconductor element and a plurality of conductor (first power-circuit-side emitter conductor plate **221**, first power-circuit-side collector conductor plate **211**, second power-circuit-side emitter conductor plate **222**, and second power-circuit-side collector conductor plate **212**) that hold the IGBT **10** therebetween and that are respectively connected to an emitter and a collector of the IGBT **10**, and a flow channel forming body **25** that houses the first and second power circuit units **201** and **202** and through which a refrigerant flows. The conductor **221** at the emitter side of the first power circuit unit **201** is disposed to face the conductor **212** at the collector side of the second power circuit unit **202**, and the conductor **221** at the emitter side of the first power circuit unit **201** and the conductor **212** at the collector side of the second power circuit unit **202** are connected to each other via a plurality of conductive cooling fins **24** which are in contact with the refrigerant. With this configuration, it is possible to provide a power conversion device that achieves both improvement in cooling performance and reduction in inductance, and increases an output.

(48) (2) The power semiconductor element includes an IGBT **10** and a diode **11**. The IGBT **10** of the first power circuit unit **201** is disposed to face the diode **11** of the second power circuit unit **202**, and the diode **11** of the first power circuit unit **201** is disposed to face the IGBT **10** of the second power circuit unit **202**. With this configuration, it is possible to reduce the inductance.

(49) (3) The flow channel forming body **25** includes an upper-surface flow channel forming body **251** that forms a flow channel on an upper surface of the first power circuit unit **201**, a lower-surface flow channel forming body **252** that forms a flow channel on a lower surface of the second power circuit unit **202**, and an intermediate flow channel forming body **253** that forms a flow channel between the first and second power circuit units **201** and **202**. With this configuration, it is possible to simultaneously cool the power circuit units **201** and **202** of the upper and lower layers while communicating the refrigerant.

(50) (4) The first and second power circuit units **201** and **202** are mounted respectively on a first board **3001** and a second board **3002** each including a conductor layer electrically connected to the conductor. With this configuration, it is possible to suppress the sizes in a height direction of the first and second power circuit units **201** and **202** disposed as the upper and lower layers.

(51) (5) The first board **3001** and the second board **3002** are configured in the same shape. With this configuration, the productivity is improved.

(52) (6) The conductor layers of the first board **3001** and the second board **3002** are disposed to face each other, and are connected respectively to a positive electrode terminal **401** and a negative electrode terminal **402** of a capacitor **40** disposed outside the flow channel forming body **25**. With this configuration, the transient current path at the switching time is shortened, and the inductance is reduced.

(53) (7) The flow channels in the upper-surface flow channel forming body **251** and the lower-surface flow channel forming body **252** are electrically connected to each other via a flow channel in the intermediate flow channel forming body **253**. With this configuration, it is possible to simultaneously cool the power circuit units **201** and **202** of the upper and lower layers.

(54) Although the present invention has been described above, an RC-IGBT may be applied to the IGBT **10**, which can further contribute to the reduction in loss of the semiconductor element for improving the fuel consumption of an HEV or an EV and to the reduction in size of the power conversion device **100**.
(55) In addition, deletion, replacement with another component, and addition of another component can be performed without departing from the technical idea of the invention, and an aspect thereof is also included in the scope of the present invention.
REFERENCE SIGNS LIST
(56) **10** IGBT **101** main electrode **102** control electrode **11** diode **12** metal bonding material **20** power module **201** first power circuit unit **202** second power circuit unit **21** collector conductor plate **211** first power-circuit-side collector conductor plate **212** second power-circuit-side collector conductor plate **22** emitter conductor plate **221** first power-circuit-side emitter conductor plate **222** second power-circuit-side emitter conductor plate **23** molding resin **24** conductive fin **25** flow channel forming body **251** upper-surface flow channel forming body **252** lower-surface flow channel forming body **253** intermediate flow channel forming body **254** refrigerant path **26** flow channel inlet **27** flow channel outlet **30** board **301***a*, **301***b* through-hole **302** via **303** power module assembly hole **3001** first board **3002** second board **31** positive electrode power supply terminal conductor **311** positive electrode power supply terminal **32** negative electrode power supply terminal conductor **321** negative electrode power supply terminal **33** AC output terminal conductor **331** AC output terminal **40** capacitor **401** positive electrode terminal **402** negative electrode terminal **50** control circuit **51** control signal wiring **52** in-flow channel board wiring **100** power conversion device

Claims

1. A power conversion device comprising: first and second power circuit units each including a power semiconductor element and a plurality of conductors that hold the power semiconductor element therebetween and that are connected to an emitter and a collector of the power semiconductor element; and a flow channel forming body which houses the first and second power circuit units and through which a refrigerant flows, wherein the conductor at an emitter side of the first power circuit unit is disposed so as to face the conductor at a collector side of the second power circuit unit, and the conductor at the emitter side of the first power circuit unit and the conductor at the collector side of the second power circuit unit are connected to each other via a plurality of conductive fins which are in contact with the refrigerant.
2. The power conversion device according to claim 1, wherein the power semiconductor element includes an IGBT and a diode, the IGBT of the first power circuit unit is disposed to face the diode of the second power circuit unit, and the diode of the first power circuit unit is disposed to face the IGBT of the second power circuit unit.
3. The power conversion device according to claim 1, wherein the flow channel forming body includes an upper-surface flow channel forming body that forms a flow channel on an upper surface of the first power circuit unit, a lower-surface flow channel forming body that forms a flow channel on a lower surface of the second power circuit unit, and an intermediate flow channel forming body that forms a flow channel between the first and second power circuit units.
4. The power conversion device according to claim 1, wherein the first and second power circuit units are mounted respectively on a first board and a second board each including a conductor layer electrically connected to the conductor.
5. The power conversion device according to claim 4, wherein the first board and the second board are configured in a same shape.
6. The power conversion device according to claim 5, wherein the conductor layers of the first board and the second board are disposed to face each other, and are connected respectively to a

positive electrode terminal and a negative electrode terminal of a capacitor disposed outside the flow channel forming body.

7. The power conversion device according to claim 3, wherein flow channels in the upper-surface flow channel forming body and the lower-surface flow channel forming body are electrically connected to each other via a flow channel in the intermediate flow channel forming body.